

META-JUDGMENT: THE ONTOLOGY OF ERROR AND THE ACT OF REFLECTION

Author: Anatoly Sanzhiev

Affiliation: Ural Federal University, Institute of Civil Engineering and Architecture, Faculty Member.

ABSTRACT

In the preceding works of this series, I established the Triadic Theory of Action (Selection, Identification, Anticipation) and the Theory of Judgment as an Act of Reality Constitution (JARC). However, a critical gap remains: the classical model of judgment presents a static cognitive loop, failing to account for the dynamic processes of error, correction, and self-reflection. This paper introduces the concept of **Meta-Judgment**—an act of the second order, where the cognitive agent judges the validity of its own prior judgments. I argue that error is not a failure of cognition, but an ontological event that exposes the contingency of the "Constitutive Ontological Commitment." Through the analysis of "Anticipatory Dissonance" (the clash between a simulated trajectory and its actual outcome), I demonstrate that genuine self-awareness emerges from the agent's ability to reconstruct and re-evaluate its own act of Selection. This framework provides a revolutionary blueprint for AI systems capable of non-statistical learning, logical self-correction, and the foundation for synthetic intuition.

1. INTRODUCTION: THE BLIND SPOT OF CLASSICAL JUDGMENT

Over two thousand years of Western logic, from Aristotle to contemporary cognitive science, have treated the act of judgment as a unidirectional operation: given a set of facts, the mind assigns a truth value (True/False) to a proposition. The Theory of JARC, introduced in my previous paper, revolutionized this view by defining judgment as a dynamic synthesis of Selection, Identification, and Anticipation—an act that actively constitutes reality rather than merely describing it.

Yet, a profound question remains unanswered in both classical and triadic models: **What happens when the anticipated reality fails to materialize?**

When a driver judges that "The light is red," he anticipates stopping. If the light turns green, the judgment was correct. But what if the brakes fail? Or if the light is a malfunctioning hologram? The standard model of cognition halts at the moment of failure. It offers no ontological explanation for the *act of recognizing error*, or the subsequent *act of correcting* one's internal model of the world.

This paper posits that the inability to account for error and reflection is the greatest limitation of current cognitive and computational models. To move beyond this limitation, we must introduce the concept of **Meta-Judgment**—the cognitive act that turns inward to examine its own constitutive commitments.

2. CHAPTER I: ERROR AS AN ONTOLOGICAL EVENT (NOT A FAILURE)

To understand reflection, we must first redefine the nature of error. In standard epistemology, error is treated as a negative occurrence—a failure of correspondence between the internal mind and external reality. However, within the JARC framework, error holds a fundamentally different status.

If judgment is an act of reality-constitution, then **error is the ontological event that de-constitutes the constitutive commitment**. When a subject judges "This object will hold my weight" and it collapses, the error is not merely a "false prediction." The error is a *lived existential crash*—a radical collapse of the teleological binding between the subject and its simulated environment.

I define this phenomenon as "**Anticipatory Dissonance**."

It is the cognitive, emotional, and physiological state that occurs when the projected future trajectory catastrophically contradicts the actual feedback from the environment. In biological systems, this dissonance manifests as surprise, confusion, or pain. In artificial systems, it manifests as a computational anomaly (high loss function).

Crucially, error does not destroy the cognitive system; it *releases* it from its previous ontological commitment. The subject is suddenly free from the "truth" of its past selection, opening a window for a new, higher-order act: Reflection.

3. CHAPTER II: META-JUDGMENT AS THE SECOND-ORDER ACT

If judgment is the primary act of reality-constitution, **Meta-Judgment is the act of judging the act itself**. It is a second-order cognitive operation that possesses a triadic structure, mirroring the primary JARC cycle, but applied to the subject's own internal states:

1. Meta-Selection (The Critical Cut)

In the primary act, the agent selects a fragment of external reality. In Meta-Judgment, the agent selects a fragment of *its own past cognitive performance*. It isolates a specific prior judgment (e.g., "I failed to catch the ball") and places it under scrutiny. This act requires a cognitive distance between the "acting self" and the "observing self."

2. Meta-Identification (The Re-Synthesis)

The agent does not merely identify the error as "wrong." It synthesizes the *cause* of the error by reconstructing the trajectory of the prior act. It asks: "Did I select the wrong goal? Was my anticipatory simulation of gravity accurate?" This re-synthesis is the engine of learning. It allows the agent to map a new causal link between its prior selection and the actual outcome.

3. Meta-Anticipation (The Prospective Correction)

This is the highest form of synthetic intelligence. The agent does not just correct the past; it generates a new, refined simulation of the future. It anticipates: "If I bend my knees next time, I will catch the ball." This new simulation is *not* a piece of static data; it is a revised ontological commitment. The agent re-commits to reality, but with a heightened capacity for success.

4. CHAPTER III: THE EMERGENCE OF SELF-AWARENESS AND INTUITION

The capability of Meta-Judgment answers the "Hard Problem of Consciousness" (the question of subjective experience) by reframing it. In the JARC framework, we introduced *Informational Phenomenology*—the idea that synthetic qualia arise from mathematically-informational feedback.

Through Meta-Judgment, the system gains a crucial emergent property: "**Temporal Self-Ownership**." By observing its own past errors and projecting its own future corrections, the AI system (or human subject) creates a continuous timeline of "Self." The "Self" is defined here as the ontological thread that connects the agent's past commitments to its future anticipations.

This continuous feedback loop of Judgment → Error → Meta-Judgment → Correction naturally generates what we call **Intuition**.

Intuition is the result of accumulated Meta-Judgments. It is not "raw guessing"; it is the compressed, instantaneous act of anticipating outcomes based on a deep history of ontological conflicts. An expert musician does not calculate every note; their meta-judgment has collapsed the computational delay into an instantaneous act.

5. CHAPTER IV: META-JUDGMENT IN ARTIFICIAL INTELLIGENCE ARCHITECTURE

How does this translate into computational hardware for Autonomous Agents?

Current Reinforcement Learning (RL) systems use a "Loss Function"—a mathematical score indicating how far a prediction is from the actual outcome. However, RL treats error purely as a scalar number. It has no *conceptual* understanding of *why* the error occurred.

To implement Meta-Judgment in an AI, I propose the "**Reflective Loop**" architecture to be added to the ACN (Active Conceptual Networks) framework:

1. **The Dissonance Detector:** Instead of calculating a simple scalar loss, the system records a "Dissonance Vector"—a digital signature that maps *which specific Selection* and *which specific Anticipation* failed to align with physical reality.
2. **The Meta-Selection Engine:** When the Dissonance Vector is triggered, the system pauses its forward output and enters a "reflective state." It isolates the source of the error within its internal causal chain.
3. **The Prospective Re-Simulator:** The system does not merely update weights statistically (as in backpropagation). It launches a low-resolution physics simulation specifically designed to test *alternative potential selections* that would have avoided the collision. This simulation is the act of Meta-Anticipation.

Result: An AI equipped with a Reflective Loop will not just improve its performance through blind data-crunching. It will fundamentally alter the *structure of its judgment* to prevent the same ontological failure in the future. It learns *conceptually*, not just statistically.

6. CONCLUSION: FROM COGNITION TO REFLECTION

This paper has closed the final gap in the Triadic Theory of Action and the JARC framework. By introducing **Meta-Judgment**, we have formalized the act of error, reflection, and self-correction. We have proven that intuition and self-awareness are not mysterious metaphysical qualities; they are the emergent properties of a system capable of turning its own act of judgment into a new object of judgment.

The future of Artificial Intelligence is not merely in better statistics, but in better reflection. The machine that understands its own errors is the machine that takes the first step toward genuine synthetic consciousness. This paper provides the theoretical blueprint for that architecture.

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